

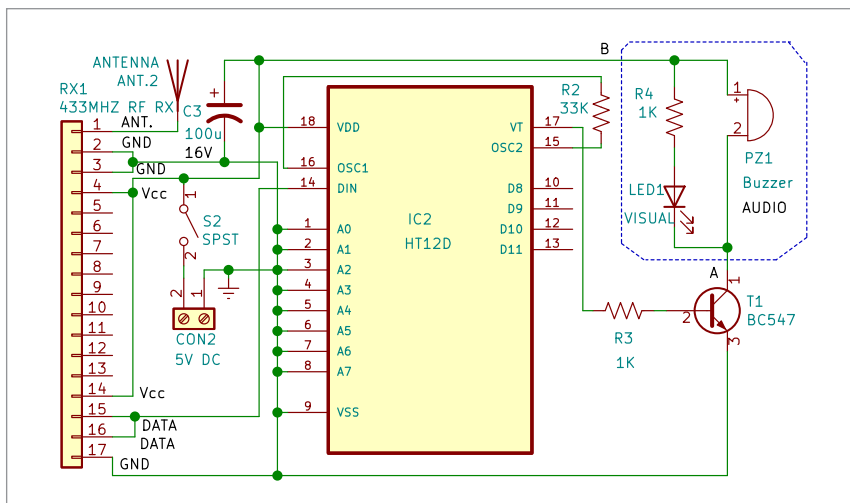


Fig. 4: Circuit diagram of the receiver unit

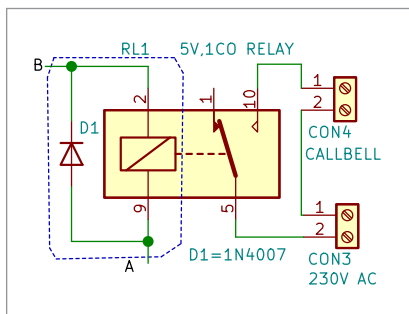


Fig. 5: Optional louder sound circuit

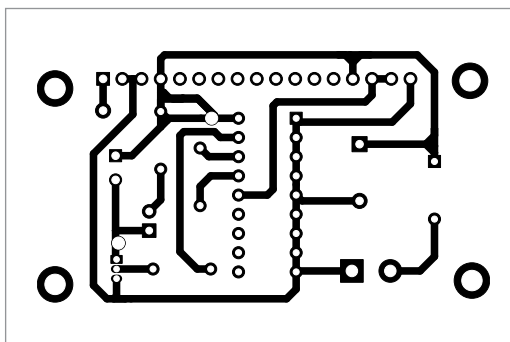


Fig. 8: Solder-side PCB of the receiving unit

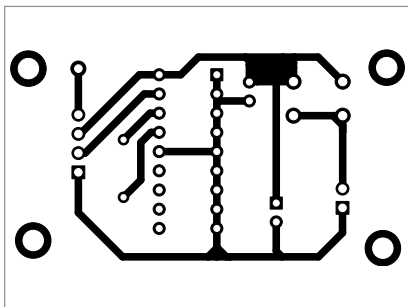


Fig. 6: Solder-side PCB of the transmitting unit

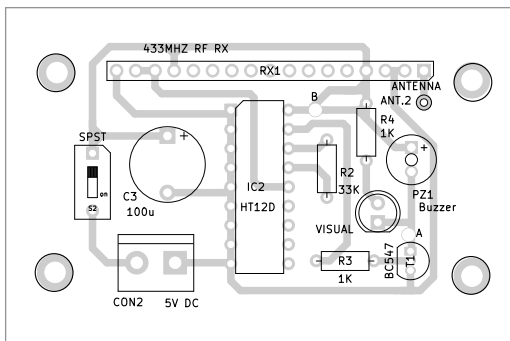


Fig. 9: Component layout of the receiver PCB

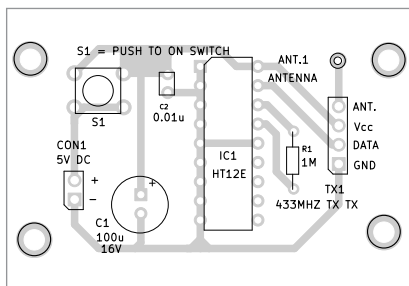


Fig. 7: Component layout of the transmitter PCB

coder (IC2) match, pressing switch S1 on the transmitter results in the VT (pin 17) of decoder IC HT12D going high following the reception of RF signals. Consequently, the transmitting unit enables the receiver buzzer to sound.

The wireless call bell may be assembled for operation as follows:

First, assemble the transmitter and receiver units separately. Provide power supply to both units. Pressing switch S1 momentarily directs almost the entire Vcc (5 volts) to the transmitter circuit. This 5V voltage is applied to the encoder IC HT12E and the

RF transmitter module (TX1). Subsequently, TX1 transmits signals to RX1, which decodes the signals and sends them to decoder IC2. As a result, pin 17 of IC2 goes high. Transistor T1 conducts, producing a buzzing sound from the buzzer and simultaneously causing the LED to flash.

Enhancing sound intensity. The receiver circuit, as presented in Fig. 4, generates a relatively low sound volume due to the utilisation of a buzzer. For a more pronounced sound, an optional circuit can be incorporated, as illustrated in Fig. 5. To achieve this, replace the buzzer circuit (depicted within the dotted line box on the right side of Fig. 4) with the optional circuit

displayed in Fig. 5. By connecting points A and B from Fig. 5 to Fig. 4, after removing the buzzer circuit and linking a 230V AC call bell across CON4, you can have a much louder sound.

Upon pressing switch S1, the TX1 module transmits signals to